Review for the PhD thesis titled “Compositional Machine Learning for Time-Verifiable, Safe, and Explainable Cyber-Physical Systems” submitted by Sobhan Chatterjee

This research thesis focuses on modern safety-critical cyber-physical systems that are designed with learning-enabled components (examples of such systems include autonomous driving and medical diagnostics). The premise of the thesis is that for such systems compositional and explainable designs are necessary to achieve high assurance and ensure verified safety. After providing a detailed introduction and background technical information in Chapter 1, the thesis makes technical contributions across three chapters. In Chapter 3, the thesis proposes techniques to decompose and transform a monolithic Machine Learning (ML) model into a compositional design for both classification as well as regression tasks. For classification, the idea is to decompose the model into separate binary classifiers for each class (except 1) and then merge the outputs using different techniques. For regression, the idea is to decompose the feature space into separate clusters and develop a model for each cluster independently. The thesis has also developed some compilation tools to transform these models automatically into VHDL code for prototyping and evaluation on the FPGA boards. In Chapter 4, the focus is only compositional safety policy synthesis, training of ML models for each sub-policy and policy-driven merging of their outputs to derive the overall system pipeline. The chapter also presents a runtime enforcement mechanism that monitors the system for policy violations in real-time and has the capability to alter the control actions to prevent such violations. Although the specific technical contributions of this chapter are not necessarily novel, it presents an interesting workflow and case-study for policy-driven compositional design of safety-critical systems. Finally, in Chapter 5, the thesis focuses on the case-study of mood score prediction for depressed individuals based on feature data gathered using smart watches as well as explicit inputs from individuals. The dataset was gathered using 14 individuals participating in a study in UCSD. The thesis proposes both a monolithic as well as a compositional (feature clustering based) design for personalized ML models addressing this problem, and evaluates them specifically for explainability using well-established metrics. Again, although the technical contributions of this chapter are not novel, the proposed workflow for model selection and evaluation are interesting and the experimental evaluations are detailed and thorough.

Overall, the thesis is well written and easy to follow. The author provides substantial technical background information in the thesis for the reader to follow the technical contributions presented therein, and I am very appreciative of this fact. The technical contributions of the thesis are interesting, specifically from a design workflow perspective and I believe the insights presented in the thesis would be valuable for safety-critical cyber-physical systems. Therefore, I am supportive of this thesis and recommend that the candidate proceed for oral defence.

I have several suggestions for improvement of the thesis. Some of the major comments are summarized below. I am also attaching an annotated version of the thesis that provides detailed comments. Any highlighted text in the annotated version w/o associated comments usually refers to typographical/grammatical errors and those should be rather apparent.

Chapter 1 (Introduction):

- I really like the Introduction chapter. However, up until Section 1.5, it reads like background material. I would suggest that you move content up to section 1.5 to a separate background Chapter that follows the introduction. In Chapter 1, it would be great if you can concisely and concretely provide motivation for your specific PhD work, define the technical problems that you worked on, and highlight the key contributions/novelty of the thesis. If you can provide this introduction w/o the above background description at a high enough level of technical abstraction, that would be great. In my view, this would generate interest among the readers for the thesis early on.

- Please expand Section 1.6 with more details. In particular, the research questions can be concretized, technically speaking. Although the high-level questions are clear, it would be helpful if you can explain with examples what these questions mean, what is the closely related literature on this and what are the research gaps technically speaking?

Chapter 2 (Literature Review):

- Everything presented prior to Section 2.3.3 reads like technical background information. Although it is well written, it does not necessarily represent literature review related to the technical focus of this thesis. I would recommend to integrate this content in a new background chapter together with some content from Chapter 1 (see comment above regarding Chapter 1).
- The premise of the thesis is that there is a general lack of research on decomposition of monolithic ML models into smaller networks. However, this claim seems far-fetched, given that the two main rival technologies currently spearheading the autonomous driving initiatives precisely claim differentiation on this specific aspect (a.k.a, Google Waymo vs. Tesla Autopilot). Waymo is considered to be resource intensive with multi-modal inputs and decomposed ML pipelines, whereas Autopilot is purely based on high-precision cameras and a monolithic end-to-end design. I would suggest that a more thorough investigation of the literature should be done to better highlights these existing technical trends.
- The use of FPGAs for implementing time-critical ML functionalities needs stronger justification (Section 2.3.4). The go-to hardware for today's autonomous driving functionality are embedded GPUs. Given this trend, how would one justify the use of FPGAs? These arguments need further strengthening, since a significant contribution of this thesis is the design of the automatic VHDL code generator.

Chapter 3 (Compositional ML for Classification and Regression):

- Section 3.7 presents results for the chapter. Although detailed results are presented, there is a lack of discussion on the intuition behind the observed results. The reader is unable to get any insights from the observed results. I would encourage the author to update this section and provide some insights to explain the why behind the results. Are the presented results (for example in Figure 3.8) specific to the considered datasets? Can we expect these results and the derived conclusions to hold for other datasets? What aspects of the chosen datasets lead to the observed results? Such insights would be very useful in interpreting the presented results.

Chapter 4 (Policy Composition and Runtime Enforcement):

- Policy decomposition may not always be feasible. It is important to discuss about the practical relevance of such decompositions. What examples do you have in mind? What if the policy for a module depends on the output of the previous module, which I believe will be a common scenario in practice. That is, a policy spans multiple ML models in a pipeline. So it is not clear to me how the proposed decomposition framework would work in that case. I assume in the proposal, each decomposed ML models produces an output with an associated outcome for the satisfaction (or not) of a decomposed policy. However, such a clean decomposition may not always be feasible and this could be elaborated upon.

- The motivation for Section 4.3.3 is unclear. Why are you learning the policy parameters based on a dataset and that too with a non-zero threshold for violations? What is the rationale? For instance, the policy predicates in Eq. 4.5 should be based on domain knowledge, no? How does data augment this domain knowledge? Also, in this learning process, how do you ensure first principles are not being violated by the learnt policies?
- In Section 4.3.4 you refer to approximate satisfaction for decomposed policies. However, this has far reaching impact on the overall system, and it is not clear how approximate satisfaction for individual decomposed policies would affect the violation probabilities for the overall system? Besides, your policy parameters are already approximate. Now if the satisfaction of the policy is also probabilistic, what is the impact on the overall system? How can this be assessed and how does one derive such decomposed guarantees methodically from the overall system requirements? Such aspects of the design process could be discussed in further detail.
- Please check the comment in the proof of Proposition 2 in Appendix B.2. I believe there is something missing the presented arguments.

Chapter 5 (Mood Score Prediction):

- My main concern with this chapter is its relevance to cyber-physical systems (cps), an application domain that is the focus of this thesis. How is mood score prediction for depressed individuals a cyber-physical system application? Typically, cps denote systems in which the cyber components work in close-coordination with the physical components to monitor and control the physical system. In my humble opinion, the application presented in this chapter is a classic data analysis problem that is conducted offline. I really don’t see any connection between this application and cps.